Exercises Week 2

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Exercise 2

Let $X(t)$ be a Gaussian zero-mean process with t $\in \mathbf{R}^d$, d = 1,2, and with Matérn covariance function.

- Let's draw a realization of one-dimensional Matérn with $\nu = 1.75$

Let's use the following discretisation
t = ih, h = 0.01, i = 0,...,100 in dimension one
And let's define the parameters as:
$[\nu, l, \sigma_f, \sigma_n] = [1.75, 0.45, 1.0, 0.1]$

$$t_i \in [0, 1] \tag{1}$$

The distance D_{ij}

$$D_{ij} = ||t_i - t_j||_2 \tag{2}$$

The Matérn covariance

$$\begin{aligned} C_v(d_{ij}) &= \sigma^2 \frac{2^{1-v}}{\Gamma(v)} \left(\frac{\sqrt{2v}d}{l} \right)^v K_v \left(\frac{\sqrt{2v}d}{l} \right), i \neq j \\ C_v(d_{ii}) &= \sigma_f^2, \forall i \in \{1, ..., n\} \end{aligned} \tag{3}$$

(Rasmussen, 2005)
The U matrix for Cholesky

$$\begin{aligned} U_{00} &= \sqrt{C_{00}} = \sigma_f \\ U_{ii} &= \sqrt{C_{ii} - \sum_{k=0}^{i-1} U_{ki}^2}, \forall i \in \{1, .., 100\} \\ U_{0j} &= \frac{C_{0j}}{U_{00}} \\ U_{ij} &= \frac{1}{U_{ii}} \left(C_{ij} - \sum_{k=0}^{i-1} U_{ki} U_{kj} \right), \text{for } j > i \end{aligned} \tag{4}$$

(Pourahmadi, 2014)

$$W_w \sim N(0, 1) \tag{5}$$

$$X_w = UW_w \tag{6}$$

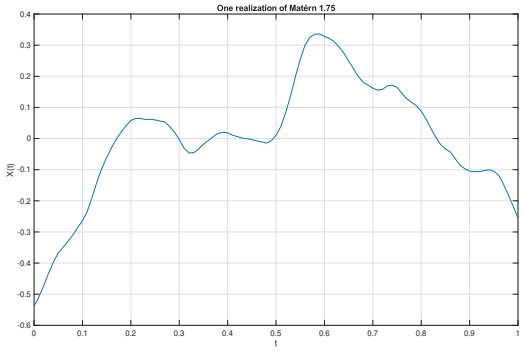
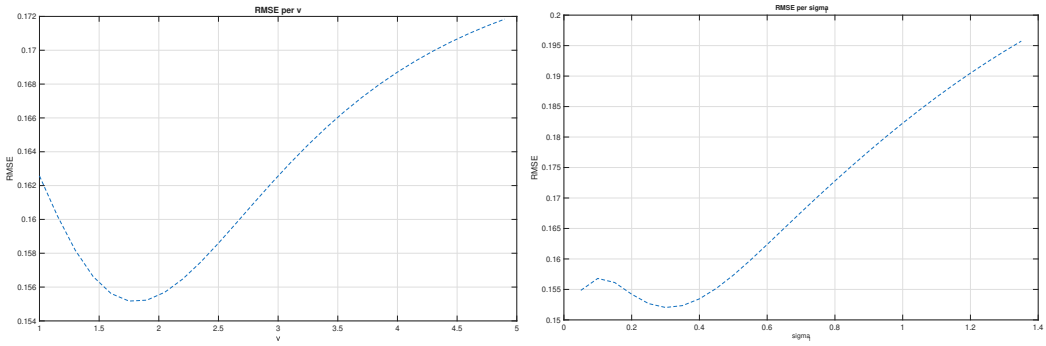


Figure 1: 1D Realization of Matérn 1.75

Now instead of having a fixed $\nu = 1.75$, let's test $\nu_k = 1 + 0.15k, k \in \{0, 1, 2, ..., 26\}$ and keep l=0.45. Then let $l_k = 0.05 + 0.05j, j \in \{0, 1, 2, ..., 26\}$ while $\nu = 1.75$ is fixed.



(a) RMSE for ν (b) RMSE for l

Figure 2: RMSE for parameters ν and l

Figure 2a shows how the RMSE values are the lowest around 1.75, after which the errors rise significantly. Figure 2b shows similar pattern though the slope of the curve is more linear. The convex shape of both figures tells us how much we can deviate from the true value while maintaining low errors.
In the case of ν, 1.75 ± 0.25 yield the lowest errors. And for l, the optimal range from 0.2 to 0.45.

Exercise 2

Next we construct a hierarchical model where we assume ν and l to be unknown and then estimate them jointly.

Let's denote $\theta = [\sigma_l, \nu]^T$. At highest level, choose the prior $p(\theta) = p(\sigma_l) \cdot p(\nu)$. Let $\sigma_l \sim U([\sigma_{min}, \sigma_{max}])$ and $\nu \sim$ Log-Normal(μ_ν, σ_ν^2).

The state x priori is $x|\theta \sim N(0, C_\theta)$.
At lowest level of the hierarchy we have the observation $y|x, \sigma_n^2 \sim N(Hx, \sigma_n^2 I)$, where H is the observation matrix.
The sampling with MCMC happens with alternating Gibb's algorithm where the state and parameters are updated each after one another based on their conditional posterior distributions.
The joint posterior is:

$$\begin{aligned} p(\theta, x|y) &\propto p(y|x) \cdot p(x|\theta) \cdot p(\theta) \\ \log p(\theta, x|y) &\propto -\frac{1}{2\sigma_n^2}||y - Hx||^2 - \frac{1}{2}x^T C_\theta^{-1} x - \frac{1}{2}\log|C_\theta| + \log p(\theta) \end{aligned} \tag{7}$$

(Gelman, 2003).
Employing Gibbs MCMC-algorithm, we express the alternating estimating as:

$$(x^{(i)}|\theta^{(i-1)}, y) -> (\theta^{(i)}, x^{(i)|y}) \tag{8}$$

At step 1, we update the state x using pCN. The proposal is expressed as:

$$x_{prop} = \sqrt{1-\beta^2} x_{curr} + \beta\zeta, \zeta \sim N(0, C_\theta) \tag{9}$$

(Chopin, 2020).
Here θ stays fixed and so the full conditional posterior is:

$$p(x|\theta, y) \propto p(y|x)p(x|\theta) \tag{10}$$

The acceptance depends on the likelihood:

$$\log a_x = -\phi(x_{prop}) + \phi(x_{curr}), \phi(x) = \frac{1}{2\sigma_n^2}||y - Hx||^2 \tag{11}$$

Next we update the parameter $\theta = [\nu, l]$ using Random Walk Metropolis-Hastings. We keep the state x fixed. First the proposal is defined as

$$\theta_{prop} = \theta_{curr} + \delta \cdot \epsilon, \epsilon \sim N(0, I) \tag{12}$$

. Here the full conditional posterior is:

$$p(\theta|x, y) = p(\theta|x) \propto p(x|\theta)p(\theta) \tag{13}$$

The acceptance rate is:

$$\log a_\theta = [\log p(x|\theta_{prop}) + \log p(\theta_{prop})] - [\log p(x|\theta_{curr}) + \log p(\theta_{curr})] \tag{14}$$

We had originally 8000 MCMC samples and we then remove the first 1000 samples (burn-in) and calculate the posterior means for the state and the parameters.

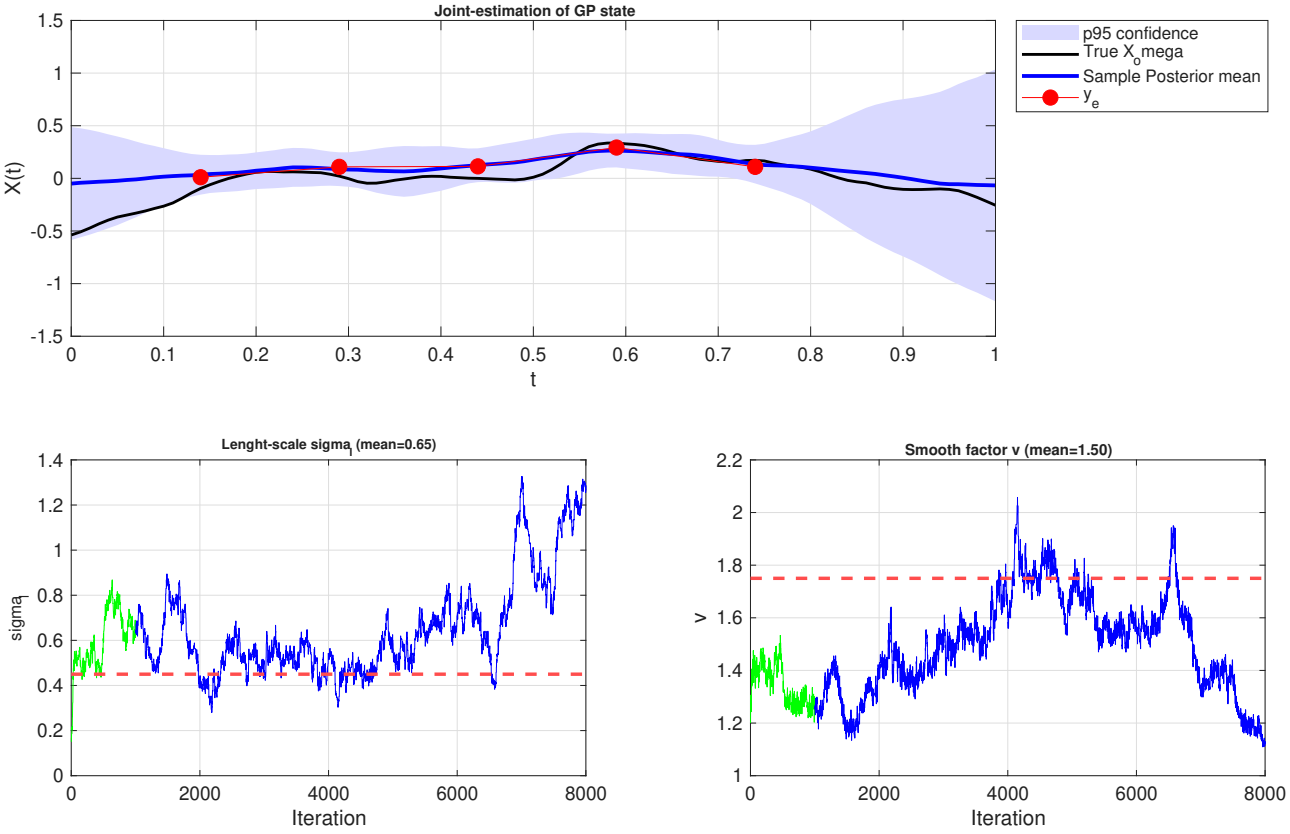


Figure 3: Joint estimation,ν and iterations, l and iterations

In Figure 3, we can see that the the uncertainty reduces at observation points and increases elsewhere. We can observe that the chains begin to drift upwards. The number of iterations is possibly too small as the consecutive samples seem correlated. For ν the estimates are slightly too high and for l, there is underestimation. For better mixing, the step size of l should revised or number of iterations augmented. The burn-in length is too short. This leads that the initial guesses still impact the chain even after the burn-in.

References

Chopin, Nicolas, P. O. (2020). *An Introduction to Sequential Monte Carlo*. Cham : Springer International Publishing.

Gelman, A. (2003). *Bayesian data analysis*. Boca Raton FL : Chapman Hall/CRC.

Pourahmadi, M. (2014). *High-Dimensional Covariance Estimation: With High-Dimensional Data*. Newark: John Wiley et Sons Ltd.

Rasmussen, Carl Edward, W. C. K. (2005). *Gaussian Processes for Machine Learning*. MIT Press.